

Zitterbewegung Cosmology: A Research Program in Ultrametric Spacetime

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Abstract

The Zitterbewegung — the rapid oscillatory motion of elementary particles predicted by the Dirac equation — is not merely a quantum mechanical artifact but the fundamental mechanism by which spacetime acquires its metric structure. This research program develops the complete framework in six pillars: (1) the Dirac equation's Zitterbewegung as mathematical foundation, (2) the correspondence between Zitterbewegung cycles and ultrametric tree-depth steps, (3) the derivation of the FLRW metric and Einstein equations from aggregate Zitterbewegung dynamics, (4) dark energy as the zero-point Zitterbewegung energy with a predicted value of $10^{-123} M_{\text{Planck}}^4$ matching observations, (5) CMB log-periodic oscillations and p-adic non-Gaussianity as observational signatures of ultrametric cosmic structure, and (6) matter-antimatter asymmetry from chirality-dependent Zitterbewegung tree-depth evolution. Testable predictions are provided for CMB, large-scale structure, neutron EDM, and neutrino experiments.

Keywords: Zitterbewegung, Dirac equation, ultrametric cosmology, dark energy, CMB, baryogenesis, p-adic cosmology

Pillar Structure

1. **Dirac Zitterbewegung — Mathematical Foundation** — Velocity operator, oscillation derivation, Foldy-Wouthuysen transformation, external fields, Compton scale
2. **Ultrametric Tree-Depth Correspondence** — Valuation map, discrete time, adelic structure, information-theoretic connection, Bekenstein bound from Zitterbewegung
3. **Spacetime Metric from Aggregate Zitterbewegung** — Stress-energy tensor, Einstein-Zitterbewegung correspondence, FLRW derivation, Friedmann equations
4. **Dark Energy as Zero-Point Zitterbewegung** — Vacuum energy sum, Planck cutoff prediction (10^{-123}), Pauli cancellation, equation of state $w = -1$, CKN bound, quintessence bridge
5. **CMB and Ultrametric Structure** — Log-periodic oscillations, p-adic non-Gaussianity, Sachs-Wolfe effect, CMB polarization parity, pre-registered test protocol

6. **Matter-Antimatter Asymmetry** — Sakharov conditions, CP violation as tree-depth, GUT/leptogenesis/electroweak baryogenesis, neutron EDM prediction

Individual pillar files available in the source repository at `zitterbewegung-pillar{1-6}.md`.